

Hasse Diagram



Draw the Hasse diagram for the following posets.

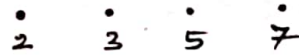
1. R is the "divides" relation applied to $A = \{1, 2, 4, 8\}$

Sol:



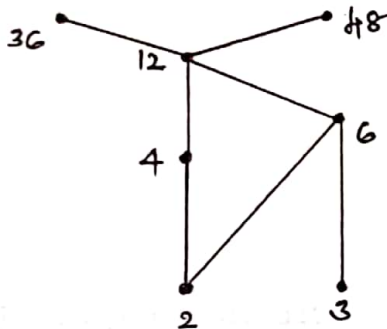
2. R is the "divides" relation applied to $A = \{2, 3, 5, 7\}$

Sol:



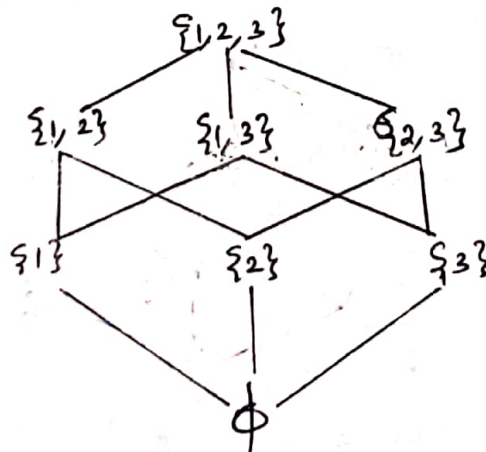
3. R is the "divides" relation on $B = \{2, 3, 4, 6, 12, 36, 48\}$

Sol:



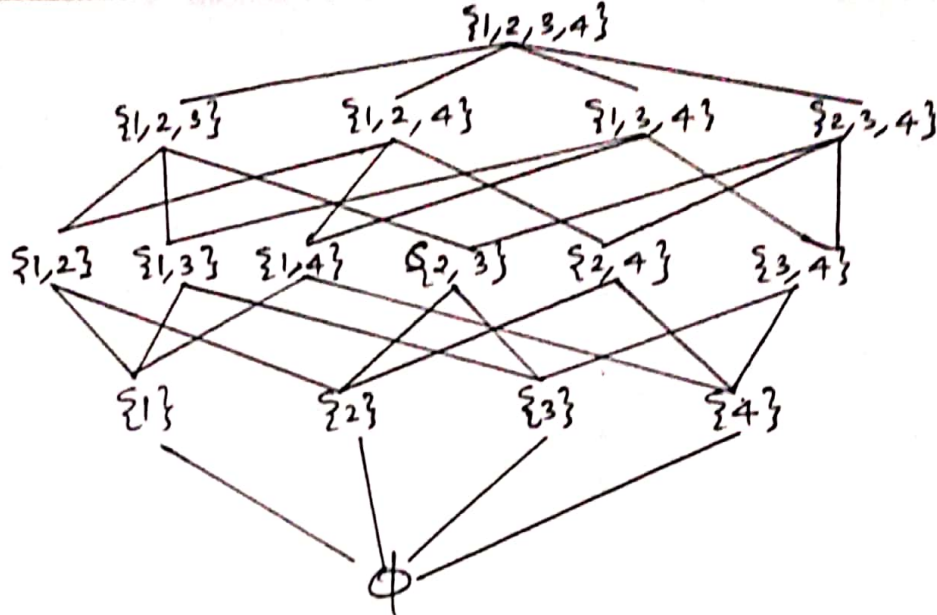
4. with $U = \{1, 2, 3\}$ and $A = P(U)$, ' R ' is the Subset relation on A .

Sol: $P(U) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$



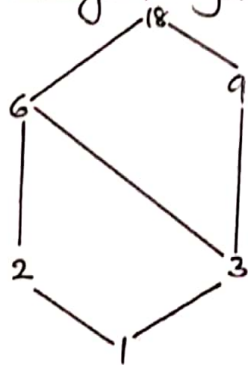
5. Draw the Hasse diagram for the poset $(P(U), \subseteq)$ where $U = \{1, 2, 3, 4\}$

Sol: $P(U) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{4\}, \{1, 2\}, \{1, 3\}, \{1, 4\}, \{2, 3\}, \{2, 4\}, \{3, 4\}, \{1, 2, 3\}, \{1, 2, 4\}, \{1, 3, 4\}, \{2, 3, 4\}, \{1, 2, 3, 4\}\}$



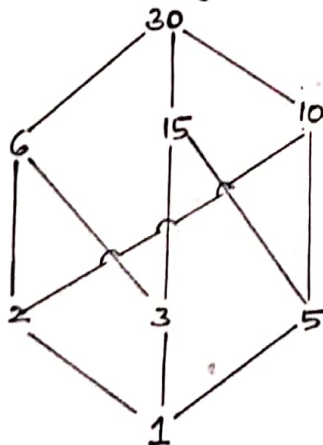
6. Let $A = \{1, 2, 3, 6, 9, 18\}$ and define R on A by xRy if $x|y$. Draw the Hasse diagram for the poset (A, R)

Sol:

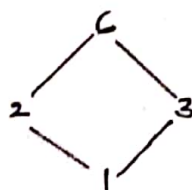


7. Let $D_{30} = \{1, 2, 3, 5, 6, 10, 15, 30\}$ and let the relation D be "divisor" on D_{30} . Draw the Hasse diagram for D_{30} with D .

Sol:

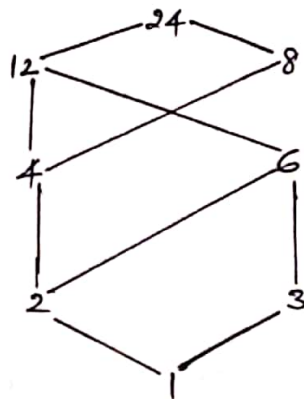


8. Draw the Hasse diagram for $(D_6, |)$.
Sol: $D_6 = \{1, 2, 3, 6\}$



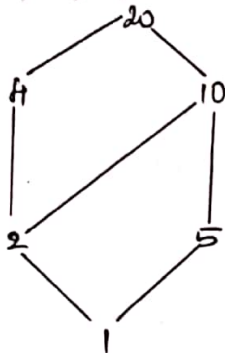
9. Draw the Hasse diagram for $(D_{24}, |)$

Sol: $D_{24} = \{1, 2, 3, 4, 6, 8, 12, 24\}$



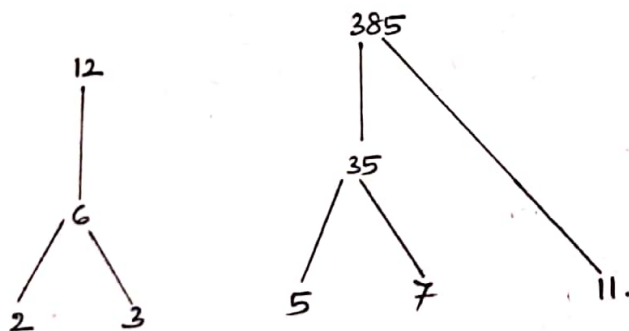
10. Draw the Hasse diagram for $(D_{20}, |)$

Sol: $D_{20} = \{1, 2, 4, 5, 10, 20\}$



11. Draw the Hasse diagram for $A = \{2, 3, 5, 6, 7, 11, 12, 35, 385\}$, where R is the "divides" relation on A .

Sol:



Maximal & Minimal elements

Let (A, R) be a poset. Then x in A is a maximal element of A if there does not exist an element $a \in A$ such that $x R a$.
Similarly, let (A, R) be a poset. Then y in A is a minimal element of A if there does not exist an element $b \in A$ such that $b R y$.



Least element & Greatest element-

Of (A, R) is a poset, then an element $x \in A$ is called a least element if xRa for all $a \in A$. An element $y \in A$ is called a greatest element if aRy for all $a \in A$.



Theorem: If the poset (A, R) has a greatest (least) element, then the element is unique.

Pf: Suppose that $x, y \in A$ and both are the greatest elements.
 Since x is the greatest element $\Rightarrow yRx$ ($y \leq x$) for $y \in A$.
 Since y is the greatest element $\Rightarrow xRy$ ($x \leq y$) for $x \in A$.
 Since ' R ' is antisymmetric $\Rightarrow x = y$.

In the same manner we can prove that least element is unique.

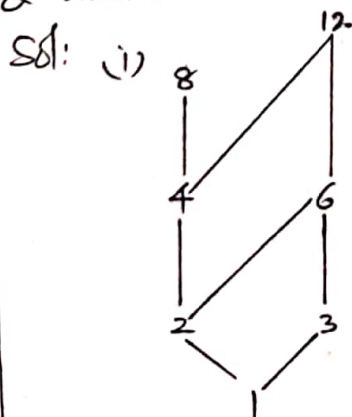


Upper bound & Lower bound

Let (A, R) is a poset with $B \subseteq A$. An element $a \in A$ is called a lower bound of B if ' aRb ' for all $b \in B$.

Likewise an element $y \in A$ is called an upper bound of B if bRy for all $b \in B$.

1. Let $A = \{1, 2, 3, 4, 6, 8, 12\}$ and R be the partial ordering on A defined by aRb if a divides b . (i) Draw the Hasse diagram of the poset (A, R) (ii) let $B = \{1, 4, 6\}$, Find the upper bounds & lower bounds on B . (iii) let $B = \{2, 4\}$, Find the upper bounds & lower bounds on B .



(ii) $B = \{1, 4, 6\}$

Lower bounds of $B = \{1\}$ (aRb)

Upper bounds of $B = \{12\}$ (bRy)

(iii) $B = \{2, 4\}$

Lower bounds of $B = \{1, 2\}$

Upper bounds of $B = \{4, 8, 12\}$

* Greatest lower bound & Least upper bound.

A element $x' \in A$ is called a greatest lower bound (glb) of B if it is a lower bound of B and if for all other lower bounds x'' of B we have $x'' R x'$.

Similarly $y' \in A$ is a least upper bound (lub) of B if it is an upper bound of B and if $y' R y''$ for all other upper bounds y'' of B .

1. Let $D_{30} = \{1, 2, 3, 5, 6, 10, 15, 30\}$ and let the relation D be the divisibility on D_{30} . Find (i) all lower bounds of 10 & 15 (ii) glb of 10 & 15 (iii) all upper bounds of 10 & 15 (iv) lub of 10 & 15.

Sol: Lower bounds of $B = \{10, 15\}$ is $\{1, 5\} \Rightarrow \text{glb}(10, 15) = 5$
Upper bounds of $B = \{10, 15\}$ is $\{30\} \Rightarrow \text{lub}(10, 15) = 30$.

2. Let $L = \{1, 3, 5, 7, 15, 21, 35, 105\}$ and let ' $\leq$ ' be the relation divides on L . Find (i) upper bounds of 3 & 7 (ii) lub of 3 & 7 (iii) lower bounds of 15 & 21 (iv) glb of 15 & 21 (v) upper bounds of 15 & 35 (vi) lub & glb of 15 & 35.

Sol: upper bounds of 3 & 7 is $\{21, 105\} \Rightarrow \text{lub}(3, 7) = 21$.
Lower bounds of 15 & 21 is $\{1, 3\} \Rightarrow \text{glb}(15, 21) = 3$.
Upper bounds of 15 & 35 is $\{105\} \Rightarrow \text{lub}(15, 35) = 105$.
Lower bounds of 15 & 35 is $\{1, 5\} \Rightarrow \text{glb}(15, 35) = 5$.

3. Let $U = \{1, 2, 3, 4\}$ with $A = P(U)$ and R be the Subset relation on A . For each of the following Subsets of B of A determine the glb & lub.

(i) $B = \{\{1\}, \{2\}\}$ (ii) $B = \{\{1\}, \{2\}, \{3\}, \{1, 2\}\}$ (iii) $B = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$
(iv) $B = \{\{1\}, \{1, 2\}, \{1, 3\}, \{1, 2, 3\}\}$ (v) $B = \{\{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}\}$.

Sol: $P(U) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{4\}, \{1, 2\}, \{1, 3\}, \{1, 4\}, \{2, 3\}, \{2, 4\}, \{3, 4\}, \{1, 2, 3\}, \{1, 2, 4\}, \{1, 3, 4\}, \{2, 3, 4\}, \{1, 2, 3, 4\}\}$

(i) Lower bounds of $B = \{\{1\}, \{2\}\}$ is $\{\emptyset\} \Rightarrow \text{glb} = \emptyset$

Upper bounds of $B = \{\{1\}, \{2\}\}$ is $\{\{1, 2\}\} \Rightarrow \text{lub} = \{1, 2\}$.

(ii) Lower bounds of $B = \{\{1\}, \{2\}, \{3\}, \{1, 2\}\}$ is $\{\emptyset\} \Rightarrow \text{glb} = \emptyset$.

Upper bounds of $B = \{\{1\}, \{2\}, \{3\}, \{1, 2\}\}$ is $\{\{1, 2, 3\}\} \Rightarrow \text{lub} = \{1, 2, 3\}$

- (iii) Lower bounds of $B = \{\emptyset, \{1\}, \{2\}, \{1,2\}\}$ is $\{\emptyset\} \Rightarrow glb = \emptyset$
 Upper bounds of $B = \{\emptyset, \{1\}, \{2\}, \{1,2\}\}$ is $\{\{1,2\}\} \Rightarrow lub = \{1,2\}$
- (iv) Lower bounds of $B = \{\{1\}, \{1,2\}, \{1,3\}, \{1,2,3\}\}$ is $\{\emptyset, \{1\}\} \Rightarrow glb = \{1\}$
 Upper bounds of $B = \{\{1\}, \{1,2\}, \{1,3\}, \{1,2,3\}\}$ is $\{\{1,2,3\}\} \Rightarrow lub = \{1,2,3\}$
- (v) Lower bounds of $B = \{\{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}\}$ is $\{\emptyset\} \Rightarrow glb = \emptyset$
 Upper bounds of $B = \{\{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}\}$ is $\{\{1,2,3\}\} \Rightarrow lub = \{1,2,3\}$

4. Consider the poset $(\mathbb{R}, \leq)$. Let $B = \{x \in \mathbb{R} ; 1 \leq x \leq 2\}$ Find lub & glb.

Sol: The glb of $1 \leq x \leq 2$ is $1 \rightarrow \min(1,2)$
 The lub of $1 \leq x \leq 2$ is $2 \rightarrow \max(1,2)$

* Lattices

A lattice is a partially order set (A, R) in which for every element of $a, b \in A$, the ^{both} glb (a, b) and lub (a, b) exists.

In other words, a lattice is a poset (A, R) in which every pair of elements has a glb and a lub.

We denote glb (a, b) by $a \wedge b$ or $a \cdot b$ and call it the meet of a & b .
 Similarly lub (a, b) by $a \vee b$ or $a + b$ and call it the join of a & b .

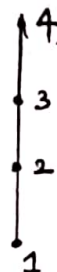
Note: The poset $(\mathbb{Z}^+, R)$ with $a R b \Rightarrow a | b$ i.e. " a divides b " is a lattice. Here $glb(a, b) = gcd(a, b) = a \wedge b$ & $lub(a, b) = lcm(a, b) = a \vee b$.

1. For each of the following posets, draw the Hasse diagram. (OR) Write the meet join table and determine whether the poset is a lattice.

(i) $(D_8, |)$

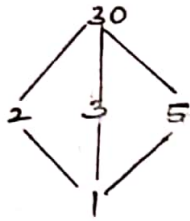
Sol: $D_8 = \{1, 2, 4, 8\}$

$\therefore (D_8, |)$ is a lattice



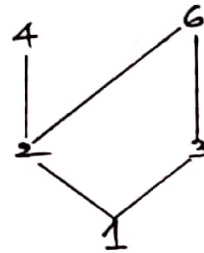
2. Draw the Hasse diagram and determine which of the following posets $(A, \mid)$ are lattices, why

(i) $A = \{1, 2, 3, 5, 30\}$



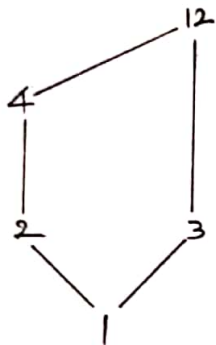
$\therefore (A, \mid)$ is a Lattice.

(ii) $A = \{1, 2, 3, 4, 6\}$



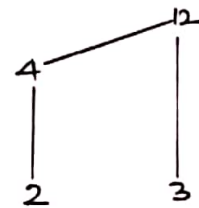
$\therefore (A, \mid)$ is not lattice, Since $\text{lub}(4, 6)$ does not exist.

(iii) $A = \{1, 2, 3, 4, 12\}$



$\therefore (A, \mid)$ is a lattice.

(iv) $A = \{2, 3, 4, 12\}$



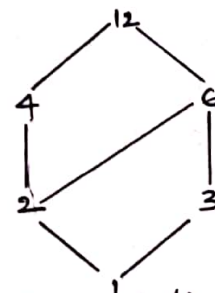
$\therefore (A, \mid)$ is not a lattice because $\text{glb}(2, 3)$ does not exist.

(v) $A = \{1, 3, 6, 30\}$



$\therefore (A, \mid)$ is a lattice

(vi) $A = \{1, 2, 3, 4, 6, 12\}$



$\therefore (A, \mid)$ is a lattice.

3. Let $A = \{1, 2, 3, 12\}$ (a) Construct the matrix table for glb & lub in $(A, \mid)$ (b) Construct the meet-join table (c) Is $(A, \mid)$ a lattice? (d) Is $\text{lub}(2, 3) = \text{lcm}(2, 3)$.

glb = gcd

Meet (OR)

$\wedge$	1	2	4	8
1	1	1	1	1
2	1	2	2	2
4	1	2	4	4
8	1	2	4	8

lub = lcm

Join

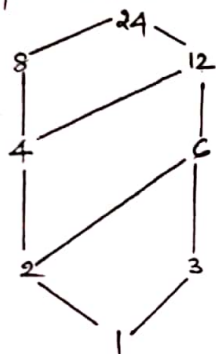
$\vee$	1	2	4	8
1	1	2	4	8
2	2	2	4	8
4	4	4	4	8
8	8	8	8	8

$\therefore (D_8, 1)$ is a lattice

(ii) $(D_6, 1)$

(iii) $(D_{24}, 1)$

Sol: $D_{24} = \{1, 2, 3, 4, 6, 8, 12, 24\}$



$\therefore (D_{24}, 1)$ is a Lattice

(OR)

glb = gcd

Meet

$\wedge$	1	2	3	4	6	8	12	24
1	1	1	1	1	1	1	1	1
2	1	2	1	2	2	2	2	2
3	1	1	3	1	3	1	3	3
4	1	2	1	4	2	4	4	4
6	1	2	3	2	6	2	6	6
8	1	2	1	4	2	8	4	8
12	1	2	3	4	6	4	12	12
24	1	2	3	4	6	8	12	24

lub = lcm

Join

$\vee$	1	2	3	4	6	8	12	24
1	1	2	3	4	6	8	12	24
2	2	2	6	4	6	8	12	24
3	3	6	3	12	6	24	12	24
4	4	4	12	4	12	24	12	24
6	6	6	6	12	6	24	12	24
8	8	8	24	24	8	24	24	24
12	12	12	12	12	12	24	12	24
24	24	24	24	24	24	24	24	24

(iv) $(D_{30}, 1)$

(v) $(D_{20}, 1)$

(vi) $(D_{35}, 1)$

(a)

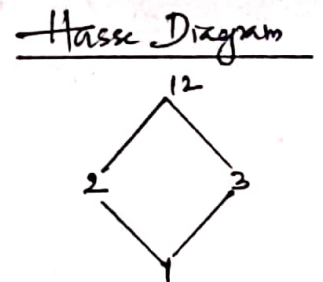
$\wedge$ glb	1	2	3	12
1	1	1	1	1
2	1	2	1	2
3	1	1	3	3
12	1	2	3	12

$\vee$ lub	1	2	3	12
1	1	2	3	12
2	2	2	(12)	12
3	3	(12)	3	12
12	12	12	12	12

(b)

Meet-Join table

$\wedge \vee$	1	2	3	12
1	1	2	3	12
2	1	2	12	12
3	1	1	3	12
12	1	2	3	12



(c) $(A, 1)$ is a lattice

(d) No, $\text{lcm}(2, 3) = 6$ but $\text{lub}(2, 3) = 12$.

4. Let $A = \{2, 12, 18, 36\}$ (a) Construct the matrix table for glb & lub (b) Construct meet join table (c) Is $(A, 1)$ a lattice (d) Is $\text{glb}(12, 18) = \text{gcd}(12, 18)$

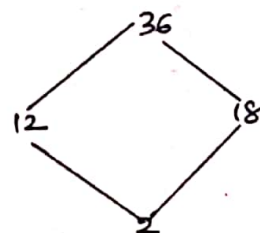
Sol: a)

(glb) $\wedge$	2	12	18	36
2	2	2	2	2
12	2	12	(2)	12
18	2	(2)	18	18
36	2	12	18	36

(lub) $\vee$	2	12	18	36
2	2	12	18	36
12	12	12	36	36
18	18	36	18	36
36	36	36	36	36

(b)

$\wedge \vee$	2	12	18	36
2	2	12	18	36
12	2	12	36	36
18	2	2	18	36
36	2	12	18	36



(d) No, $\text{gcd}(12, 18) = 6$ but $\text{glb}(12, 18) = 2$.

- Note: 1) The poset $(P(S), \subseteq)$. Here $\text{glb}(a, b) = a \cap b$ and $\text{lub}(a, b) = a \cup b$.
- 2) The poset $(A, |)$. Here $\text{glb}(a, b) = \text{gcd}(a, b)$ and $\text{lub}(a, b) = \text{lcm}(a, b)$.
- 3) The poset $(A, \leq)$. Here $\text{glb}(a, b) = \min(a, b)$ and $\text{lub}(a, b) = \max(a, b)$.

eg: 1) $(\mathbb{N}, R)$ $x R y \Rightarrow x \leq y$ where $\mathbb{N} = \{1, 2, 3, 4, 5, 6, \dots\}$

Let $B = \{4, 5\}$.

$\Rightarrow$ Lower bounds of $B = \{1, 2, 3, 4\}$ & upper bound = $\{5, 6, 7, 8, \dots\}$

$\Rightarrow \text{glb}(B) = \min(4, 5) = 4$ & $\text{lub} = \max(4, 5) = 5$.

2) $(\mathbb{Z}^+, R)$ $a R b \Rightarrow a | b$. where $\mathbb{Z}^+ = \{1, 2, 3, 4, 5, 6, \dots\}$

Let $B = \{4, 5\}$

$\Rightarrow$ Lower bounds of $B = \{1\}$

& upper bound = $\{20, 40, 60, 80, 100, \dots\}$

$\Rightarrow \text{glb}(B) = \text{gcd}(4, 5) = 1$.

$\text{lub}(B) = \text{lcm}(4, 5) = 20$.

3) $(P(S), \subseteq)$, $S = \{1, 2, 3\}$, $P(S) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$

Let $B = \{2\}$ & $C = \{2, 3\}$

$\Rightarrow$ Lower bounds of $(B, C) = \{\emptyset, \{2\}\}$ & upper bounds of $(B, C) = \{\{2, 3\}, \{1, 2, 3\}\}$

$\Rightarrow \text{glb}(B, C) = \{2\}$

$\Rightarrow \text{lub}(B, C) = \{2, 3\}$

4. Let $S = \{1, 2, 3\}$, then P.T the poset $(P(S), \subseteq)$ is a lattice.

Sol: Given $S = \{1, 2, 3\} \Rightarrow P(S) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$

$\wedge$		$\emptyset$	$\{1\}$	$\{2\}$	$\{3\}$	$\{1, 2\}$	$\{1, 3\}$	$\{2, 3\}$	$\{1, 2, 3\}$
$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$
$\{1\}$	$\emptyset$	$\{1\}$	$\emptyset$	$\emptyset$	$\{1\}$	$\{1\}$	$\emptyset$	$\{1\}$	$\{1\}$
$\{2\}$	$\emptyset$	$\emptyset$	$\{2\}$	$\emptyset$	$\{2\}$	$\emptyset$	$\{2\}$	$\{2\}$	$\{2\}$
$\{3\}$	$\emptyset$	$\emptyset$	$\emptyset$	$\{3\}$	$\emptyset$	$\{3\}$	$\{3\}$	$\{3\}$	$\{3\}$
$\{1, 2\}$	$\emptyset$	$\{1\}$	$\{2\}$	$\emptyset$	$\{1, 2\}$	$\{1\}$	$\{2\}$	$\{1, 2\}$	$\{1, 2\}$
$\{1, 3\}$	$\emptyset$	$\{1\}$	$\emptyset$	$\{3\}$	$\{1\}$	$\{1, 3\}$	$\{3\}$	$\{1, 3\}$	$\{1, 3\}$
$\{2, 3\}$	$\emptyset$	$\emptyset$	$\{2\}$	$\{3\}$	$\{2\}$	$\{3\}$	$\{2, 3\}$	$\{2, 3\}$	$\{2, 3\}$
$\{1, 2, 3\}$	$\emptyset$	$\{1\}$	$\{2\}$	$\{3\}$	$\{1, 2\}$	$\{1, 3\}$	$\{2, 3\}$	$\{1, 2, 3\}$	$\{1, 2, 3\}$

Sub U	Joint							
	$\emptyset$	$\{1\}$	$\{2\}$	$\{3\}$	$\{1,2\}$	$\{1,3\}$	$\{2,3\}$	$\{1,2,3\}$
$\emptyset$	$\emptyset$	$\{1\}$	$\{2\}$	$\{3\}$	$\{1,2\}$	$\{1,3\}$	$\{2,3\}$	$\{1,2,3\}$
$\{1\}$	$\{1\}$	$\{1\}$	$\{1,2\}$	$\{1,3\}$	$\{1,2\}$	$\{1,3\}$	$\{1,2,3\}$	$\{1,2,3\}$
$\{2\}$	$\{2\}$	$\{1,2\}$	$\{2\}$	$\{2,3\}$	$\{1,2\}$	$\{1,2,3\}$	$\{2,3\}$	$\{1,2,3\}$
$\{3\}$	$\{3\}$	$\{1,3\}$	$\{2,3\}$	$\{3\}$	$\{1,2,3\}$	$\{1,3\}$	$\{2,3\}$	$\{1,2,3\}$
$\{1,2\}$	$\{1,2\}$	$\{1,2\}$	$\{1,2\}$	$\{1,2,3\}$	$\{1,2\}$	$\{1,2,3\}$	$\{1,2,3\}$	$\{1,2,3\}$
$\{1,3\}$	$\{1,3\}$	$\{1,3\}$	$\{1,2,3\}$	$\{1,3\}$	$\{1,2,3\}$	$\{1,3\}$	$\{1,2,3\}$	$\{1,2,3\}$
$\{2,3\}$	$\{2,3\}$	$\{1,2,3\}$	$\{2,3\}$	$\{2,3\}$	$\{1,2,3\}$	$\{1,2,3\}$	$\{2,3\}$	$\{1,2,3\}$
$\{1,2,3\}$	$\{1,2,3\}$	$\{1,2,3\}$	$\{1,2,3\}$	$\{1,2,3\}$	$\{1,2,3\}$	$\{1,2,3\}$	$\{1,2,3\}$	$\{1,2,3\}$

$\therefore (P(S), \subseteq)$ is a lattice.

* Sublattice

Let (A, R) be a lattice. A non-empty subset B of A is called a Sublattice of A if for any $a, b \in B$, $a \vee b$ and $a \wedge b \in B$.

Note: Let $(D_n, |)$ be a lattice and 'm' be any integer such that $m|n$ then all possible D_m 's are Sublattices of $(D_n, |)$. Also all multiples of m are Sublattices.

1. Consider the Lattices $(P(S), \subseteq)$ where $S = \{1, 2, 3\}$. Determine whether or not each of the following is a Sublattice of L

i) $A = \{\emptyset, \{1, 2\}, \{2, 3\}, \{1, 2, 3\}\}$

ii) $B = \{\emptyset, \{1\}, \{1, 2\}, \{1, 2, 3\}\}$

iii) $C = \{\emptyset, \{3\}, \{1, 3\}, \{1, 2, 3\}\}$

(Ans: it is a Sublattice)

iv) $D = \{\{1\}, \{3\}, \{1, 3\}, \{1, 2, 3\}\}$

(Ans: it is not lattice)

v) $E = \{\emptyset, \{3\}, \{1, 2\}, \{1, 2, 3\}\}$

(Ans: it is a Sublattice)

Sol: (i)

$n=n$	$\emptyset$	$\{1, 2\}$	$\{2, 3\}$	$\{1, 2, 3\}$
$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$
$\{1, 2\}$	$\emptyset$	$\{1, 2\}$	$\{2\}$	$\{1, 2, 3\}$
$\{2, 3\}$	$\emptyset$	$\{2\}$	$\{2\}$	$\{2, 3\}$
$\{1, 2, 3\}$	$\emptyset$	$\{1, 2\}$	$\{2, 3\}$	$\{1, 2, 3\}$

it is not a Sublattice

Since $\text{glb}\{\{2, 3\}, \{1, 2\}\} = \{2\}$ does not exist in A .

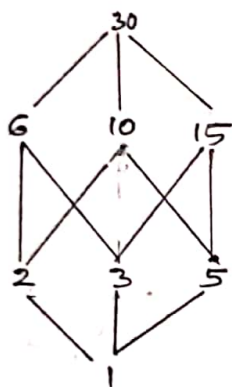
$$(ii) B = \{\emptyset, \{1\}, \{1,2\}, \{1,2,3\}\}$$

glb $\wedge$	$\{1\}$	$\{1,2\}$	$\{1,2,3\}$	lub $\vee$	$\emptyset$	$\{1\}$	$\{1,2\}$	$\{1,2,3\}$
$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\{1\}$	$\{1,2\}$	$\{1,2,3\}$
$\{1\}$	$\emptyset$	$\{1\}$	$\{1\}$	$\{1\}$	$\{1\}$	$\{1\}$	$\{1,2\}$	$\{1,2,3\}$
$\{1,2\}$	$\emptyset$	$\{1\}$	$\{1,2\}$	$\{1,2\}$	$\{1,2\}$	$\{1,2\}$	$\{1,2\}$	$\{1,2,3\}$
$\{1,2,3\}$	$\emptyset$	$\{1\}$	$\{1,2\}$	$\{1,2,3\}$	$\{1,2,3\}$	$\{1,2,3\}$	$\{1,2,3\}$	$\{1,2,3\}$

$\therefore$ it is a sublattice

2. Determine the sublattices of D_{30} that contains at least 4 elements.

Sol: we know that $D_{30} = \{1, 2, 3, 5, 6, 10, 15, 30\}$



The sublattices of D_{30} that contain at least four elements are

- (i) $\{1, 2, 6, 30\}$
- (ii) $\{1, 3, 6, 30\}$
- (iii) $\{1, 3, 15, 30\}$
- (iv) $\{1, 5, 10, 30\}$
- (v) $\{1, 5, 15, 30\}$.

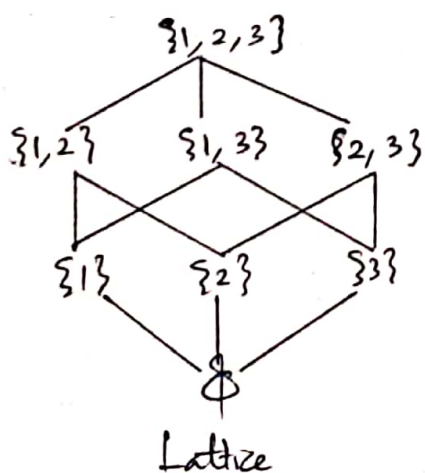


Dual or Reversed lattice

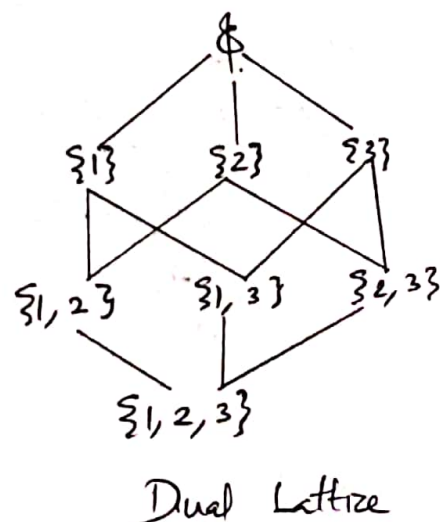
If $(A, \leq) = [A, \vee, \wedge]$ is a lattice, then $(A, \geq) = [A, \wedge, \vee]$ is known as dual or reversed lattice, which is obtained by interchanging glb & lub.

eg:- The reversed lattice of $(P(S), \subseteq)$ is $(P(S), \supseteq)$

For $S = \{1, 2, 3\}$



$\Rightarrow$



* Some Special lattices

1. Complete lattice : A lattice is called Complete if each of its non-empty subsets have a lub and a glb. The greatest element of a lattice is denoted by I and the least element of a lattice is denoted by O . Every complete lattice must have I and O .
eg: Every finite lattice is Complete, $(D_{20}, \mid)$ is Complete.
2. Bounded lattice : A lattice is said to be bounded if it has a greatest element I and a least element O . The elements I and O are known as bounds of the lattice.
eg: The $(P(S), \subseteq)$ is a bounded lattice since it has $I = S$ & $O = \emptyset$.
Note: For a bounded lattice an element b is said to be a complement of a if $a \cdot b = O = \text{glb}(a, b)$ and $a + b = I = \text{lub}(a, b)$
3. Complement lattice : A lattice is said to be complement lattice if every element has atleast one complement.
eg: In $B = \{1, 5, 7, 35\}$, here greatest element = 35 & least element = 1.
 $\text{glb}(1, 35) = 1$ & $\text{lub}(1, 35) = 35 \Rightarrow$ the complement of 1 is 35.
 $\text{glb}(5, 7) = 1$ & $\text{lub}(5, 7) = 35 \Rightarrow$ the complement of 5 is 7.
 But in $D_{20} = \{1, 2, 4, 5, 10, 20\}$ greatest element = 20 & least element = 1.
 $\text{glb}(1, 20) = 1$ & $\text{lub}(1, 20) = 20 \Rightarrow$ the complement of 1 is 20
 $\text{glb}(2, 10) = 2 \neq 1$ & $\text{lub}(2, 10) = 10 \neq 20 \Rightarrow 2$ & 10 have no complement
 $\text{glb}(4, 5) = 1$ & $\text{lub}(4, 5) = 20 \Rightarrow$ the complement of 4 is 5.
4. Distributive lattice : A lattice $(A, \cdot, +)$ is said to be a distributive lattice if for any $a, b, c \in A$
 $a + (b \cdot c) = (a + b) \cdot (a + c)$ ie $+$ is distributive over $\cdot$.
 & $a \cdot (b + c) = (a \cdot b) + (a \cdot c)$ ie $\cdot$ is distributive over $+$.
 In other words $+$ & $\cdot$ are distributive over each other in a distributive lattice.

1. Show that the lattice $(\mathbb{Z}^+, \leq)$ is distributive

Sol: Under the partial order $\leq$, $\text{glb}(a, b) = \min(a, b)$ & $\text{lub}(a, b) = \max(a, b)$.

Let $a, b, c \in \mathbb{Z}^+$

$$\begin{aligned} a \cdot (b+c) &= \text{glb}(a, b+c) \\ &= \min(a, b+c) \\ &= \min(a, \text{lub}(b, c)) \\ &= \min(a, \max(b, c)) \\ &= \max(\min(a, b), \min(a, c)) \\ &= \max(\text{glb}(a, b), \text{glb}(a, c)) \\ &= \max(a \cdot b, a \cdot c) \\ &= \text{lub}(a \cdot b, a \cdot c) = (a \cdot b) + (a \cdot c) \end{aligned}$$

$$\begin{aligned} (a+b) \cdot c &= \text{lub}(a, b \cdot c) \\ &= \max(a, \text{glb}(b, c)) \\ &= \max(a, \min(b, c)) \\ &= \min(\max(a, b), \max(a, c)) \\ &= \min(\text{lub}(a, b), \text{lub}(a, c)) \\ &= \min(a+b, a+c) \\ &= \text{glb}(a+b, a+c) = (a+b) \cdot (a+c) \end{aligned}$$

2. Let $A = \{1, 2, 3, 5, 30\}$ s.t. $(A, |)$ is a lattice by constructing a meet-join table 2) P.T. ' $\cdot$ ' is not distributive over ' $+$ ' in this lattice by identifying elements a, b, c in A for which $a \cdot (b+c) \neq (a \cdot b) + (a \cdot c)$. 3) P.T. ' $+$ ' is not distributive over ' $\cdot$ ' by showing $a + (b \cdot c) \neq (a+b) \cdot (a+c)$ for some elements in A .

Sol: $\text{glb} = \wedge$

	1	2	3	5	30
1	1	1	1	1	1
2	1	2	1	1	2
3	1	1	3	1	3
5	1	1	1	5	5
30	1	1	3	5	30

$\text{lub} = \vee$

	1	2	3	5	30
1	1	2	3	5	30
2	2	2	(30)	(30)	30
3	3	(30)	3	(30)	30
5	5	10	(30)	5	30
30	30	30	30	30	30

$\therefore (A, |)$ is a lattice.

2) Let $a=2$, $b=3$, $c=5$. Then

$$2 \cdot (3+5) = \text{glb}(2, 3+5) = \text{glb}(2, \text{lcm}(3,5)) = \text{glb}(2, 30) = 2.$$

$$(2 \cdot 3) + (2 \cdot 5) = \text{glb}(2,3) + \text{glb}(2,5) = 1 + 1 = \text{lub}(1,1) = 1.$$

$\therefore 2 \cdot (3+5) \neq (2 \cdot 3) + (2 \cdot 5) \Rightarrow \cdot$ is not distributive over $+$.

3) Let $a=2$, $b=3$, $c=5$.

$$2 + (3 \cdot 5) = \text{lub}(2, 3 \cdot 5) = \text{lub}(2, \text{glb}(3,5)) = \text{lub}(2, 1) = 2.$$

$$(2+3) \cdot (2+5) = \text{lub}(2,3) \cdot \text{lub}(2,5) = 30 \cdot 30 = \text{glb}(30,30) = 30.$$

$\therefore 2 + (3 \cdot 5) \neq (2+3) \cdot (2+5) \Rightarrow +$ is not distributive over $\cdot$.

3. Show that the lattice $(A, |)$, where $A = \{1, 2, 3, 4, 12\}$ is not distributive, $\therefore +$ is not distributive over $\cdot$ and vice versa.

4. Find the sublattices of D_{24} that contains at least five elements.

5. Find the Complements for the following lattices $(A, |)$

i) $A = \{1, 2, 3, 6\}$

ii) $A = \{1, 2, 3, 4, 12\}$

iii) $A = \{1, 2, 3, 5, 30\}$